EXCEL APPLICATIONS

Functions & Formular

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Open excel, create data sets for ample applications of the following:

1. **Describe cell referencing types in excel with examples**

In Excel, **cell referencing** determines how a cell address behaves when copied or moved. There are three main types of cell referencing:

1. Relative Cell Reference

- **Definition**: Changes based on the position where it is copied or moved.
- **Example**:
  - Formula in **B2**: `=A2+10`
  - If copied to **B3**, it changes to `=A3+10`.

2. Absolute Cell Reference

- **Definition**: Remains fixed when copied or moved by using `$` symbols before the row and column.
- **Example**:
  - Formula in **B2**: `=$A$1+10`
  - If copied to **B3**, it **remains** `=$A$1+10`.

3. Mixed Cell Reference

- **Definition**: Either the row or column is fixed while the other part changes.
- **Types**:
  - **Fixed column, changing row** (`$A1`)
  - **Fixed row, changing column** (`A$1`)
- **Example**:
  - Formula in **B2**: `=$A2+10`
  - If copied to **C3**, it changes to `=$A3+10` (column remains **A**, row changes).

These referencing methods are useful in formulas, especially when working with large datasets.

2. **HOW TO Sorting & filtering data IN EXCEL WITH EXAMPLES**

Sorting & Filtering Data in Excel

Sorting and filtering are essential functions in Excel that help organize and analyze data effectively.

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1. Sorting Data in Excel

Sorting allows you to arrange data in ascending or descending order based on one or more columns.

Steps to Sort Data:

1. Select the range of data you want to sort (e.g., `A1:C10`).
2. Go to the **"Data"** tab.
3. Click **"Sort"**.
4. Choose the column you want to sort by.
5. Select either **Ascending (A → Z) or Descending (Z → A)**.
6. Click **OK**.

Example:

Name	Age	Salary
John	25	5000
Alice	30	7000
Bob	22	4000

- **Sorting by Age (Ascending):**
  - Bob → John → Alice
- **Sorting by Salary (Descending):**
  - Alice → John → Bob

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2. Filtering Data in Excel

Filtering allows you to display only specific data while hiding the rest.

Steps to Apply a Filter:

1. Select the dataset.
2. Go to the **"Data"** tab.
3. Click **"Filter"** (a small drop-down arrow appears on each column header).
4. Click the arrow in the column header and select the criteria (e.g., filter Age > 25).
5. Click **OK** to apply the filter.

Example:

Name	Age	Salary
John	25	5000
Alice	30	7000
Bob	22	4000

- **Filter Age > 25:**
  - Only **Alice** appears.
- **Filter Salary ≥ 5000:**
  - **John and Alice** appear.

Advanced Sorting & Filtering:

- **Sort by multiple columns:** Sort first by "Salary" and then by "Age."
- **Custom Filters:** Use options like "Greater than," "Less than," or "Contains."
- **Clear Filters:** Click **Data → Clear Filter** to remove filters.

Sorting and filtering make it easier to manage large datasets and analyze information efficiently.

3. **HOW TO Logical, Text, Date & Time IN EXCEL WITH EXAMPLES**

Logical, Text, and Date & Time Functions in Excel with Examples

Excel provides various functions to handle **logical operations**, **text manipulation**, and **date & time calculations** efficiently.

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1. Logical Functions in Excel

Logical functions return **TRUE** or **FALSE** based on conditions.

Common Logical Functions:

Function	Description	Example
`IF`	Returns one value if condition is TRUE, another if FALSE	`=IF(A1>50, "Pass", "Fail")`
`AND`	Returns TRUE if all conditions are TRUE	`=AND(A1>50, B1<100)`

Function	Description	Example
OR	Returns TRUE if at least one condition is TRUE	=OR(A1>50, B1<100)
NOT	Reverses a logical value	=NOT(A1>50)

Example:

Score	Result Formula
60	=IF(A1>50, "Pass", "Fail") → **Pass**
40	=IF(A2>50, "Pass", "Fail") → **Fail**

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2. Text Functions in Excel

Text functions help manipulate text strings.

Common Text Functions:

Function	Description	Example
LEFT	Extracts a specific number of characters from the start	=LEFT("Excel",2) → **"Ex"**
RIGHT	Extracts a specific number of characters from the end	=RIGHT("Excel",2) → **"el"**
MID	Extracts a substring from the middle	=MID("Excel",2,2) → **"xc"**
LEN	Returns the length of a text string	=LEN("Excel") → **5**
CONCATENATE / TEXTJOIN	Combines multiple text strings	=TEXTJOIN(" ",TRUE,A1,B1)

Example:

Name	Formula	Result
John	=LEFT(A1,2)	**"Jo"**

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3. Date & Time Functions in Excel

Date and time functions allow calculations with dates and times.

Common Date & Time Functions:

Function	Description	Example
`TODAY`	Returns the current date	`=TODAY()`
`NOW`	Returns the current date & time	`=NOW()`
`YEAR`	Extracts the year from a date	`=YEAR("12/03/2023")` → **2023**
`MONTH`	Extracts the month from a date	`=MONTH("12/03/2023")` → **3**
`DAY`	Extracts the day from a date	`=DAY("12/03/2023")` → **12**
`DATEDIF`	Calculates the difference between two dates	`=DATEDIF(A1, B1, "Y")` (years)
`TEXT`	Formats date/time into text	`=TEXT(A1, "DD-MMM-YYYY")`

Example:

Date	Formula	Result
12/03/2023	`=YEAR(A1)`	**2023**

These functions help in **decision-making, text processing, and date calculations**, making Excel a powerful tool for data handling!

4. **HOW TO sum, average, rounddown, roundup, sumif, averageif IN EXCEL WITH EXAMPLES**

Mathematical & Statistical Functions in Excel with Examples

Excel provides powerful functions for calculations, including **SUM, AVERAGE, ROUNDDOWN, ROUNDUP, SUMIF, and AVERAGEIF**.

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1. SUM Function

The `SUM` function adds up numbers in a range.

Syntax:

```
=SUM(range)
```

Example:

A	B
10	
20	
30	

- **Formula:** `=SUM(A1:A3)`
- **Result:** `60`

---

2. AVERAGE Function

The `AVERAGE` function calculates the mean (average) of a range.

Syntax:

```
=AVERAGE(range)
```

Example:

A	B
10	
20	
30	

- **Formula:** `=AVERAGE(A1:A3)`
- **Result:** `20`

3. ROUNDDOWN Function

The `ROUNDDOWN` function rounds a number **down** to a specified number of decimal places.

Syntax:

```
=ROUNDDOWN(number, num_digits)
```

Example:

- **Formula:** `=ROUNDDOWN(15.789, 1)`
- **Result:** `15.7`

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4. ROUNDUP Function

The `ROUNDUP` function rounds a number **up** to a specified number of decimal places.

Syntax:

```
=ROUNDUP(number, num_digits)
```

Example:

- **Formula:** `=ROUNDUP(15.789, 1)`
- **Result:** `15.8`

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5. SUMIF Function

The `SUMIF` function adds values based on a condition.

Syntax:

```
=SUMIF(range, criteria, [sum_range])
```

Example:

	A	B
10	A	

A	B
20	B
30	A

- **Formula:** `=SUMIF(B1:B3, "A", A1:A3)`
- **Result:** `10 + 30 = 40`

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6. AVERAGEIF Function

The `AVERAGEIF` function calculates the average based on a condition.

Syntax:

```
=AVERAGEIF(range, criteria, [average_range])
```

Example:

A	B
10	A
20	B
30	A

- **Formula:** `=AVERAGEIF(B1:B3, "A", A1:A3)`
- **Result:** `(10 + 30) / 2 = 20`

These functions **simplify calculations** in Excel, making it easier to handle large datasets efficiently.

5. **HOW TO min, max, count, stdev, countif, countblank, rank IN EXCEL WITH EXAMPLES**

Statistical & Counting Functions in Excel with Examples

Excel provides powerful functions for finding **minimum, maximum, count, standard deviation, and ranking values** in datasets.

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1. MIN Function

The `MIN` function returns the smallest number in a range.

Syntax:

```
=MIN(range)
```

Example:

A
10
20
30

- **Formula:** `=MIN(A1:A3)`
- **Result:** `10`

---

2. MAX Function

The `MAX` function returns the largest number in a range.

Syntax:

```
=MAX(range)
```

Example:

A
10
20
30

- **Formula:** `=MAX(A1:A3)`
- **Result:** `30`

---

3. COUNT Function

The `COUNT` function counts the number of numeric values in a range.

Syntax:

```
=COUNT(range)
```

Example:

A
10
20
Text
30

- **Formula:** `=COUNT(A1:A4)`
- **Result:** 3 (Ignores text)

---

4. STDEV Function

The `STDEV` function calculates the standard deviation of a dataset (spread of numbers from the mean).

Syntax:

```
=STDEV(range)
```

Example:

A
10
20
30

- **Formula:** `=STDEV(A1:A3)`
- **Result:** `10`

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5. COUNTIF Function

The `COUNTIF` function counts values based on a specific condition.

Syntax:

```
=COUNTIF(range, criteria)
```

Example:

A
Apple
Banana
Apple

- **Formula:** `=COUNTIF(A1:A3, "Apple")`
- **Result:** `2`

---

6. COUNTBLANK Function

The `COUNTBLANK` function counts the number of empty (blank) cells.

Syntax:

```
=COUNTBLANK(range)
```

Example:

A
10
30

- **Formula:** `=COUNTBLANK(A1:A4)`

- **Result:** 2

---

7. RANK Function

The `RANK` function assigns a rank to a number in a dataset.

Syntax:

```
=RANK(number, range, [order])
```

- **`order = 0` (Descending, highest value is rank 1)**
- **`order = 1` (Ascending, lowest value is rank 1)**

Example:

A
50
30
80

- **Formula:** `=RANK(A1, A1:A3, 0)`
- **Result:** 2 (50 is the second highest)

Summary of Functions:

Function	Purpose	Example Formula	Result
`MIN`	Smallest number in range	`=MIN(A1:A3)`	10
`MAX`	Largest number in range	`=MAX(A1:A3)`	30
`COUNT`	Counts numbers only	`=COUNT(A1:A4)`	3
`STDEV`	Standard deviation	`=STDEV(A1:A3)`	10
`COUNTIF`	Count values matching a condition	`=COUNTIF(A1:A3, "Apple")`	2
`COUNTBLANK`	Count empty cells	`=COUNTBLANK(A1:A4)`	2

Function	Purpose	Example Formula	Result
`RANK`	Rank a value in a dataset	`=RANK(A1, A1:A3, 0)`	2

These functions help in data **analysis, filtering, and ranking** in Excel!

6. HOW TO fv, pv, pmt IN EXCEL WITH EXAMPLES

Financial Functions in Excel: FV, PV, and PMT with Examples

Excel provides built-in **financial functions** to calculate **Future Value (FV), Present Value (PV), and Payment (PMT)** for financial analysis.

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1. FV (Future Value) Function

The `FV` function calculates the future value of an investment based on periodic, constant payments and a constant interest rate.

Syntax:

```
=FV(rate, nper, pmt, [pv], [type])
```

- **rate** → Interest rate per period
- **nper** → Total number of payment periods
- **pmt** → Payment per period (if applicable)
- **pv** (optional) → Present value (initial investment)
- **type** (optional) → `0` (end of period) or `1` (beginning of period)

Example:

You invest **$1000** at an **annual interest rate of 5%** for **5 years** with no additional payments.

- **Formula:** `=FV(5%/12, 5*12, 0, -1000, 0)`
- **Result:** `$1,276.45`

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2. PV (Present Value) Function

The `PV` function calculates the present value of an investment based on periodic payments and a discount rate.

Syntax:

```
=PV(rate, nper, pmt, [fv], [type])
```

- **rate** → Interest rate per period
- **nper** → Total number of payment periods
- **pmt** → Payment per period
- **fv** (optional) → Future value
- **type** (optional) → `0` (end of period) or `1` (beginning of period)

Example:

You want to know how much you need to invest today to get **$5000** in **5 years** at an **8% annual interest rate**.

- **Formula:** `=PV(8%/12, 5*12, 0, 5000, 0)`
- **Result:** `-3,395.30` (You need to invest $3,395.30 today)

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3. PMT (Payment) Function

The `PMT` function calculates the fixed payment required to repay a loan or investment.

Syntax:

```
=PMT(rate, nper, pv, [fv], [type])
```

- **rate** → Interest rate per period
- **nper** → Total number of payments
- **pv** → Loan or investment amount
- **fv** (optional) → Future value (default is `0`)
- **type** (optional) → `0` (end of period) or `1` (beginning of period)

Example:

You take a **loan of $10,000** at an **annual interest rate of 6%** for **5 years**, and you need to calculate the monthly payment.

- **Formula:** `=PMT(6%/12, 5*12, -10000, 0, 0)`
- **Result:** `$-193.33` (You must pay $193.33 per month)

Comparison of FV, PV, and PMT Functions:

Function	Purpose	Example
`FV`	Calculates future value of an investment	`=FV(5%/12, 60, 0, -1000, 0)` → `$1,276.45`
`PV`	Calculates present value needed for future value	`=PV(8%/12, 60, 0, 5000, 0)` → `3,395.30`
`PMT`	Calculates loan payments	`=PMT(6%/12, 60, -10000, 0, 0)` → `$-193.33`

These functions help in **loan calculations, investments, and financial planning** in Excel!